

1 Remote sensing large-wood storage downstream of reservoirs during
2 and after dam removal: Elwha River, Washington, USA

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Abstract

Large wood is an integral part of many rivers, often defining river-corridor morphology and habitat, but its occurrence, magnitude, and evolution in a river system are much less well understood than the sedimentary and hydraulic components. We measured large wood and alluvial sediment deposits from high-resolution orthoimages to explore the dynamics of large wood in two reaches of the Elwha River, Washington, USA, between 2012 and 2017 as it adjusted to upstream dam removals. The dataset consists of a time series of orthoimages (12.5-cm resolution) constructed using Structure-from-Motion photogrammetry on imagery from 14 aerial surveys. We map sediment, water, vegetation, and large wood in the active channel margin using automated deep-learning (DL) based image segmentation. These fully reproducible methods and model result in a maximum of 15% error between observed and estimated total wood areas and wood deposit size-distributions over the full spatio-temporal extent of the data. Areal extent of wood in the channel margin approximately doubled in the years following dam removal, with greatest increases in large wood in wider, lower-gradient sections. Large wood deposition increased between the start of dam removal (2011) and winter 2013, then plateaued. Sediment bars continued to grow up until 2016/17, assisted by a partially static wood framework deposited predominantly during the period up to winter 2013. Our datasets and DL models are made available to stimulate further studies of fluvial morphodynamics resulting from interactions between wood, sediment, vegetation, and the channel form.

Plain language summary

We measure the large wood in the Elwha River, Washington, USA, during and after dam removal. The presence of two dams had previously limited the movement of sediment and wood through the system. The removal of those dams liberated large amounts of sediment and wood from the former reservoir bottoms, which traveled downstream and deposited in the river channel. We develop an Artificial Intelligence (AI) model to measure all wood and sediment in the Elwha River corridor downstream of the two former dams, from a time-series of high-resolution imagery collected from aircraft. These measurements, accurate to within 15% of true values, provide a unique

opportunity to understand how large wood occurs and behaves over multiple years and tens of kilometers. We found that the deposition of large wood on bars was coincident with and promoted the growth of sediment bars. The AI model we made could be powerful enough to find large wood in other places and images for similar purposes.

1. Introduction

1.1. Large wood in rivers

The occurrence and movement of large wood (downed, dead pieces greater than 10 cm in diameter and 1 m long; Wohl et al., 2019a) can significantly influence the evolution of river ecosystems through bio-geomorphic feedbacks (Ruiz-Villanueva et al., 2016; Wohl et al., 2019b). In recent decades, a growing literature has documented the beneficial effects of large wood in rivers, revealing the importance of the roles that large wood plays in creating in-stream habitats (Frissell et al., 1986; Fetherston et al., 1995; Abbe and Montgomery, 1996), longitudinal and lateral (i.e., river-floodplain) connectivity (Beechie et al., 2006; Wohl et al., 2019a; Swanson et al., 2021), channel braiding and avulsion (Wohl et al., 2019a; Swanson et al., 2021), bio-geomorphic interactions (Wohl et al., 2019b), and nutrient cycling (Collins et al., 2012), among others, summarized by Wohl et al. (2017). In addition, the importance of large wood in river restoration, dam removal, fisheries, fish passage, spawning and rearing habitat, among other important aspects of applied river science, have been the subject of review by Gurnell et al. (2002), Abbe et al. (2003), Nagayama and Nakamura (2010), Wohl (2017, 2020), Wohl et al., (2019), and Swanson et al., (2021).

Wohl et al. (2019a) conceptualized the influence of interactions among water, sediment, and wood on river corridors. Emergent properties of these interactions result in a physical habitat “template” that includes the amount, diversity, and quality of available physical habitats, the connectivity of the river channel with its floodplain, dissolved nutrient availability, and water quality (Collins et al., 2012). In settings such as the U.S. Pacific Northwest, the physical habitat template is strongly influenced by the

presence of mobile and stored wood (e.g., Fetherston et al., 1995; Abbe and Montgomery, 1996; Beechie et al., 2006; Scott and Wohl 2018).

Depending on context, wood may be stored within the active channel for a varying amount of time, from less than 1 year to more than 10,000 years (e.g., Nanson et al. 1995; Wohl, 2017). Stored wood influences geomorphic processes such as sediment transport, deposition, and storage, and the formation of morphologies such as steps, pools, bars, and secondary channels (e.g., Abbe and Montgomery, 1996; Wohl et al., 2017; Grabowski and Wohl, 2021). As channel complexity increases, we expect more wood deposition, and vice versa, with hypothesized feedback processes involved (Wohl et al., 2017). At the scale of long reaches, large wood is thought to suppress sediment transport and promote sediment deposition through increased hydraulic roughness (Davidson and Eaton, 2013; Wohl, 2017), and to promote hydraulic and topographic complexity, which can promote the storage of organic matter (Pfeiffer and Wohl, 2018). For example, persistent jams of large wood are understood to trap sediment to the point that they have even been responsible for the conversion of bedrock reaches to alluvial reaches (Montgomery et al., 1996; Massong and Montgomery, 2000).

Humans have extensively modified wood recruitment (the process of wood entering an active channel or floodplain), and the abundance, sizes, mobility and spatial distributions in river channels during the modern industrial and post-industrial ages. Large wood in river channels has been a harvested resource of opportunity or removed by humans because of concerns about flooding, fish passage, and/or riverbank erosion (Wohl, 2019; Moulin et al., 2004; Abbe and Montgomery, 1996; Comiti et al., 2016). Additionally, the sources of wood from old-growth forests have been eliminated from many watersheds from logging, and dam construction across rivers worldwide may reduce wood transport downstream because of wood impoundment and removal and hydromodifications to downstream flow regimes (Comiti et al., 2008; Wohl, 2011). This widespread loss of wood has led to incidences of channel incision, including the transformation of alluvial channels into bedrock channels (Stock et al., 2005; Schanz et al., 2018). The loss of longitudinal connectivity with respect to large wood has also

created detrimental changes in coastal, nearshore, and even deep-sea environments (Wohl and Iskin, 2021).

Because of the general reduction of wood in many rivers over time, restoration efforts often include an enhancement of wood abundance to increase channel complexity and improve habitat (Warren and Kraft, 2008; Roni et al., 2015; Grabowski et al., 2019). Dam removal has the potential to enhance downstream wood and sediment storage, but studies of these changes – especially of wood – are rare (Major et al., 2017). In one prior study, large wood was documented to play a role in reservoir erosion and channel evolution after the Merrimack Village Dam was removed on the Souhegan River, New Hampshire (Pearson et al., 2011). McDowell and Hassan (2024) also report on channel changes that resulted from the removal of an upstream culvert and subsequent large sediment pulse. Changes were found to be exacerbated by the presence of a logjam that formed when the culvert was removed.

Given the importance of large wood in river ecology, geomorphology and hydrology, a better understanding of the physical, ecological, and engineering role of large wood will contribute to both better natural resource management and better hazard risk assessment and mitigation (Abbe and Brooks, 2011; Wohl et al., 2016). To achieve this, measurements of the size, abundance, persistence, and geomorphic setting of large wood are needed, over both large scales of time and distance within river corridors. These kinds of measurements, especially if paired with efforts by the large wood research community towards standardization of methods and packaging of accessible software, will allow for comprehensive assessments of wood storage patterns and persistence, which Wohl et al. (2017) and Swanson et al. (2021) identify as a major knowledge gap in riverine wood science. Such data might help better realize management objectives such as defining “target wood regimes” (Wohl et al., 2019a), which will help balance desired geomorphic and ecological characteristics with large wood-related management and hazard-mitigation considerations for any given system. Additionally, repeatable measurements of large wood abundance and distribution will inform efforts to quantitatively assess the role of wood in river form and functions (Wohl et al., 2019a; Wohl., 2020).

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135 *1.2. Remote-sensing approach to large-wood monitoring*

136 Of particular interest to the studies of wood in riverine and coastal systems are
137 repeatable methods that can be applied to measure wood and other channel
138 components at large spatial scales and over long time series. A viable alternative to field
139 work to amass these measurements, especially at catchment scales, is the use of aerial
140 remote sensing to quantify wood visible at the surface (Sendrowski and Wohl, 2021).
141 One advantage of a remote-sensing approach is its potential to capture a synoptic
142 measurement of all wood in a system at a single moment in time, including wood that
143 may be relatively inaccessible by foot or vehicle. Given the decimeter-to-meter scale of
144 some smaller wood accumulations, lidar or imagery with voxel or pixel spatial footprints
145 much less than 1-m must typically be used. This precludes the use of most of the publicly
146 available satellite imagery, which typically consists of pixels with a spatial footprint
147 larger than 1-m, for quantifying wood accumulations of all sizes. Close-range cameras
148 with an oblique vantage have been used for small sub-reach-scale studies of large wood
149 dynamics (e.g., Bertoldi et al., 2013), but are limited in spatial extent. Aerial remote
150 sensing of wood presence and transport in whole river corridors using lidar (e.g.,
151 Bertoldi et al., 2013; Atha and Dietrich, 2016; Kuiper et al., 2023), 3D point clouds from
152 Structure from Motion (SfM) photogrammetry (e.g., Spreitzer et al., 2019), or nadir-
153 perspective aerial imagery or ‘orthoimagery’ (e.g. Haschenburger and Rice 2004;
154 Smikrud and Prakash, 2006; Lassetre et al., 2008; Atha, 2014; Taminga et al., 2015;
155 Kramer et al., 2017; Sanhueza et al., 2019; Tsunetaka et al., 2021; Iroumé et al., 2023;
156 Sendrowski et al., 2023; Iroumé et al. 2023) are all promising and relatively new means
157 to investigate the spatiotemporal properties and longevity of wood storage at the scale of
158 the river corridor over multiple years (Ruiz-Villanueva et al., 2016).

159 Imagery from a wide range of aerial platforms provide opportunities to observe and
160 quantify large wood in river systems over a spectrum of spatial and temporal scales
161 (Table 1). Prior measurements from imagery have generally used manual digitization of
162 wood accumulations or manually supervised image processing workflows, rather than
163 fully automated means. Out of 14 prior studies, nine used entirely manual wood

detection, and five used semi-automated image processing methods for wood detection (Table 1). These methods require manual specification of parameters, image feature representations, and/or manual investigation of appropriate image features to extract and use for wood detection. One prior method, that of Sendrowski et al. (2023), used Deep Learning (DL) to automate the process of wood detection. In this work Sendrowski et al. (2023) showed that so-called ‘binary’ image-segmentation, whereby image pixels are classified as wood or not wood, could effectively be used within a supervised DL framework at a large scale and from high-resolution remotely sensed imagery. Manual work is also required for DL models such as Sendrowski et al. (2023), however, specifically in the development of the data required for model training, but once trained to provide measurements at sufficient accuracy and verified that the model works well over the requisite spatial and temporal scale, no additional manual effort should be necessary. Development of such models remains a challenge, but once completed, a trained model instance such as that of Sendrowski et al. (2023), when made available to other researchers, may transfer well to the task of recognizing similar large wood deposits in similar imagery of other alluvial systems, which has been demonstrated of DL models in other similar contexts (Kentsch et al. 2020; Vandaele et al. 2021). A transferable model such as this may be deemed useful as-is, upon completion of a straightforward verification based on manual digitized measurements. Alternatively, a modest amount of fine-tuning of the model may be carried out based on new labeled data. Both contexts are referred to in the DL literature as Transfer Learning, an umbrella term of strategies designed to maximize the utility of trained model instances with minimal amounts of new labeled data, often owing to the considerable effort required to train DL models from scratch, both in terms of large datasets required, and potentially significant model training time.

In the case of repeated remote sensing of a study area, a ‘datacube’ approach to large wood monitoring, consisting of spatially and temporally explicit analyses of a time-series of co-located imagery, will facilitate measurements of wood presence, absence, and persistence for every pixel location in the dataset, which can be aggregated into larger spatial scales for subsequent analysis (Gray et al., 1997 Lewis et al., 2017). A datacube of large wood will also contain, upon appropriate analysis, information about

the spatial distribution of wood storage, and the covariation between wood storage and other important elements, such as alluvial sediment, topographic elevation, channel and floodplain dimensions and morphological characteristics, or vegetation. At the location and scale of a single pixel, the occurrence and frequency of transitions between landcovers and landforms may yield information on specific processes such as recruitment and re-vegetation, and wood deposition and erosion. This approach will allow for the assessment of spatial distributions of wood deposition, such as concentrations (logjams) or dispersed pieces (i.e., individual logs) (e.g., Kraft et al. 2011), or the temporal aspects of large wood, such as persistence and analysis of transition. As such, the datacube approach to remote sensing analyses will allow for assessments of spatial and temporal patterns of large wood and may be informative in studying flow divergence, bar sedimentation processes, and other geomorphic patterns and processes.

1.3. Objectives

In this contribution we seek to develop a DL model to map large wood, alluvial sediment, water, and vegetation in an actively evolving river channel using a time-series of high-resolution orthoimagery. With the resulting time-series data, we will quantify the magnitude and persistence (residence time) of the storage term of the time-varying wood and sediment budgets, as well as an analysis of transition among wood, sediment, water, and vegetation, which reveals the specific location of, or general likelihood of, temporal transitions that are the result of physical processes. By focusing on a river undergoing significant and rapid changes because of an alteration of its wood and sediment regime by dam removal, we provide novel observations of the time-dependent effects of a large dam removal on the downstream fluvial geomorphology.

We adopt a similar methodology to Sendrowski et al. (2023), which utilizes a binary (wood / no wood) DL approach to pixelwise classification, however, we use our models to detect not just large wood, but also sediment, water, and vegetation. We compare and evaluate the use of four separate binary models for each class, as well as a single

multiclass model that simultaneously predicts all four classes. Additionally, we have made our study reproducible by making our model training codes, trained models, training data, and codes used to generate and analyze the resulting maps, available in the hope that they may facilitate river corridor mapping from high-resolution imagery acquired in other river systems (cf. Hsu et al., 2015). By our estimation, no prior large wood study involving detection from imagery (Table 1) is computationally reproducible either because: (i) the wood was detected manually, and there is no access to raw files, nor, in the majority of cases, a published schema that was employed to determine the presence/absence of large wood; (ii) the semi-automated processing scripts, if used, were not published and documented or are otherwise no longer available; or (iii) trained DL models and training data were not made publicly available.

We demonstrate that, as a monitoring tool applied to a time-series of orthoimagery alone, and without the need for supporting datasets or in situ measurements, the DL method developed here allows for accurate measurements of wood storage fluctuations and other spatial-temporal patterns resulting from dam removal. Further, high-resolution photogrammetry of wood magnitude, persistence, and transport are combined with measurements of sediment to collectively reveal a comprehensive picture of the coevolution of wood and sediment in the alluvial channel.

2. Study Site

We focus on two reaches of the lower Elwha River, which is situated on the Olympic Peninsula in western Washington, United States, and drains into the Strait of Juan de Fuca (Figure 1). Two large dams were built on the Elwha River during the early 20th century and removed nearly a century later as part of an ecosystem restoration project to reconnect anadromous fish with fluvial spawning habitats that were inaccessible because of the dams (McHenry et al., 1998; Pess et al., 2008; Duda et al., 2011). The forests that formerly occupied the site of the former Lake Mills and Lake Aldwell sites were logged prior to dam closure and reservoir filling, and the trees removed, but the root systems remained. The 32-m high Elwha Dam was constructed between 1911-1914

at ~7.4-rkm (river kilometers from the outlet) and resulted in the reservoir Lake Aldwell, whereas the 64-m high Glines Canyon Dam was constructed at ~21.6-rkm in 1927, impounding the Lake Mills reservoir. The lower Elwha Dam was fully removed between 2011 and 2012, and upper Glines Canyon Dam was removed incrementally between 2011 and 2014 (Ritchie et al., 2018a). The removal of these two dams resulted in the exposure of ~30-Mt of sediment and wood deposited in the two reservoirs, two thirds of which (~20-Mt) subsequently eroded and was transported downstream by natural fluvial processes (Ritchie et al., 2018a). The restoration project reconnected the upper watershed, steep mountainous terrain of Olympic National Park, to the lower river and coastal sea, thereby allowing passage of upper watershed sediment and wood to downstream reaches (Duda et al., 2011; Warrick et al., 2015; East et al., 2018; Ritchie et al., 2018a).

We use the river reach definitions of East et al. (2015, 2018), which define the ‘middle reach’ to be between the two dam sites, and the ‘lower reach’ to be downstream of the lower dam site (Figure 1). In our study, as described in Section 3, orthoimagery was publicly available for the two reaches downstream of former dams, but orthoimagery of the 9-km of former reservoir beds and deltas was not. The middle and lower reaches are ~8- and ~7-km in length, respectively, and the two reaches differ in terms of slope, elevation, sinuosity, and sediment conveyance (East et al., 2015, 2018). In the middle reach, the average gradient of the river channel varies between 0.007 and 0.009 at the sub-reach scale before entering the former Lake Aldwell delta and can be steeper in short canyon sections (East et al., 2018; Figure 1c). The lower reach, in contrast, is flatter with average gradients ranging between 0.004 to 0.007 with a broader active channel width and a low gradient river outlet region (~0.0013) that is just outside of our area of analysis (Figure 1c, d; Warrick et al., 2011; East et al., 2015). The geomorphology, sedimentology, and geological history of the Elwha River basin have been reviewed and described elsewhere (McHenry et al., 1998; Pohl, 2004; Duda et al. 2008; Pess et al., 2008; Latterell et al. 2008; Warrick et al., 2011, 2015; Draut et al., 2011; East et al., 2015, 2018; Ritchie et al., 2018a).

Removal of the two dams resulted in the mobilization of ~20 Mt of sediment (roughly two-thirds of the total sediment deposited in the reservoirs) to the coastal ocean over five years (2011–2016) (Ritchie et al., 2018a). Warrick et al (2015), Ritchie et al. (2018a) and East et al. (2018) and Bountry et al. (2018) observed an increase in wood downstream and in the coastal zone, respectively, following the removal of the two dams, but did not quantify wood occurrence. Logjam area was suggested to increase following dam removal, however, due to both the increase in large wood flux from upstream observed at locations assumed to be representative, or from more comprehensive, but spatially limited, measurements (Warrick et al., 2015; DeMott, 2021), as well as qualitative evidence that more local redistribution of wood (including some recruitment of floodplain trees) as the higher sediment load in the river led to increased lateral channel mobility and growth of sediment bars (East et al., 2015, 2018; Ritchie et al., 2018a). This increased influx and deposition of large wood followed an era of wood reduction in the Elwha River during the 20th century from wood sequestration in the two reservoirs, but also logjam removal, floodplain logging, and channelization of the river mainstem documented by McHenry et al. (1998), Duda et al. (2008), and Draut et al. (2011). In response to these historic reductions of large wood in the Elwha River, engineered logjams (ELJs) were built in the lower Elwha River by the Lower Elwha Klallam Tribe and partners between 1999 and 2010 to enhance floodplain connectivity, restore juvenile salmonid habitats including scour pools that provide relatively cool water refugia, and mitigate bank erosion (McHenry et al., 2007; Pess et al., 2012). As shown below, we track the time-history of both non-ELJ and ELJ large wood in the Elwha River.

This history of large wood abundance in the Elwha River catchment is similar to other steep, mountainous rivers of the Pacific Northwest (PNW) region as documented by several studies (e.g., Abbe and Montgomery 1996, 2003; Montgomery et al., 1996; Haschenburger and Rice, 2004; Athar, 2014; Scott and Wohl, 2016). Historic reductions in wood abundance and distribution due to damming of lowland rivers, “...appear to have fundamentally changed the morphology, dynamics, and habitat abundance and characteristics of lowland rivers” (Collins et al., 2002) in the PNW region. Additionally, wood currently entering PNW rivers is generally smaller than historic log diameters and

lengths, since most old growth forests have been logged in the region (Montgomery et al 1996). The overall reduction in wood influx and size coupled with wood harvesting from channels has led to fewer, smaller logjams in most rivers in this region (McHenry et al., 1998). Lastly, ELJs have been widely used in the PNW in an attempt to decrease riverbank erosion and improve aquatic habitats (Abbe et al., 1997; Abbe and et al., 2003; Nagayama and Nakamura, 2010), where they may have had roles in the geomorphic evolution of these channels, including the recruitment of large wood into a mobile subpopulation (Wohl et al., 2023), or a subpopulation that undergoes intermittent transport and storage, as well as affecting river flow efficiency (McHenry et al., 2007) if logjams are sufficiently large to create backwater effects. Thus, the Elwha River case study provides a unique opportunity to examine the supply, magnitude, timing, and persistence of wood under unusually high wood and sediment loading within a geographic region in which large wood plays a central role in the ecologic, hydrologic, and geomorphic setting.

Qualitative assessments of wood resulting from the, to date, world's largest dam removal project on the Elwha River suggested that wood distributions and abundances changed markedly along the downstream river channel, within the floodplain deposits, and along the shoreline near the river outlet following dam removal (Warrick et al., 2015; Draut and Ritchie, 2015; Shafroth et al., 2016; Ritchie et al., 2018a). An initial geospatial analysis indicated that the area of large-wood deposits had grown 8-fold between 2011 and 2016 in the Elwha River downstream of both dam sites (Bountry et al., 2018). Tracking of individually tagged pieces of large wood confirmed that wood traveled from both upstream reservoirs to the channel reach immediately downstream of both dams after their respective removals (Leung et al., 2019). While that study focused its field component on measuring sediment scour in the former reservoirs above the river reaches considered in this study, laboratory experiments were also conducted by Leung et al. (2019) that shed light on the important role rapid wood deposition likely had in scouring pools in the gravel-dominated Elwha River system. These observations were corroborated by the hand digitization of logs and log jams along the middle reach of the Elwha River channel between former Glines Canyon Dam and former Lake Aldwell, in a quasi-annual time-series of orthoimagery, beginning August 2012 and

ending February 2020 by DeMott (2021). That analysis revealed that the total number of individual logs initially increased by approximately four-fold during the first year after dam removal (i.e., 2012 - 2016) before returning to background levels, while the total log jam area along the middle reach of the Elwha River increased between 2012 and 2020. While conclusive, these hand-based measurements only provide information for one reach of the river at an almost-annual frequency, and they could not provide total areal extents of the large wood from the entire active channel. Neither can these measurements provide information about the influences of large wood on channel morphodynamics, because the measurements result in information about large wood only, and not the evolving distributions of sediment, water, vegetation, and other landcovers and landforms. In addition, manual digitization of wood in imagery does not result in a computationally reproducible method, nor the development of an automated workflow that might be transferable to other contexts and studies.

3. Data and Methods

Our methods include the development of a Deep Learning (DL) approach to detect not just large wood, but also sediment, water, and vegetation. We utilize a series of 14 published orthophotos from the middle and lower reaches of the Elwha River to detect large wood over a relatively large spatial extent (a total of 246-ha of active river channel, or 3444-ha over 14 surveys). This results in a greater temporal frequency than most studies, with 2.5 surveys per year (by our estimation, in studies comparable to ours, only Iroumé et al. (2023) report a higher temporal frequency of 4.85 surveys per year; Table 1).

3.1. Orthoimagery

To investigate the spatial and temporal evolution of large wood storage in the Elwha River, we relied on a series of aerial imagery collected as part of interdisciplinary before-after/control-impact studies of dam-removal response (Ritchie et al., 2018a, b). The

orthoimagery data from Ritchie et al. (2018b) consist of 14 orthoimages of the middle and lower reaches collected between 2012-04-07 and 2017-09-22 (Figure 1). The imagery represents a small (and already published) subset of data products from over 100 aerial surveys conducted between 2012-03-22 and present-day (the interested reader is referred to Supplemental Figure S1). Only the 14 images utilized here are already published and publicly available at the time of writing. A subset of our goals in this paper are to develop and present methods that result in a trained model that would be applicable to the entire 100+ survey record.

The orthoimagery was generated using SfM photogrammetry (Warrick et al., 2017; Spreitzer et al., 2019; Over et al., 2021) using a digital camera mounted in a small airplane wing. This time series of imagery from the two reaches downstream of the dam sites has previously been used by Ritchie et al. (2018a) and East et al. (2015, 2018) in geomorphic assessments of river-corridor changes. Example imagery from this time-series for a small representative section of the middle reach is shown in Figure 2, revealing changes in the areal distributions of both large wood and sediment. A complete time series of this specific site in the Elwha middle reach is provided in the Supplemental Information (Figure S2), which provides examples of the appearance of the channel under the varying light and weather conditions captured in the data set. Coincident and concurrent Digital Elevation Models (DEMs) of sufficient quality for fine-scale analysis are not available for this study, so our objective is to develop a technique for measuring the 2D accumulations of wood and sediment as apparent in high-resolution geospatial imagery alone, with no supporting elevation data or in-situ measurements.

The 0.125 x 0.125-m pixel resolution imagery were reprojected to be in the NAD83(2011) / UTM zone 10N projection. We note that our orthoimagery are slightly larger than the 0.1-m pixel resolution threshold suggested by Ruiz-Villanueva et al. (2016) to accurately measure all large wood pieces. While this threshold was determined by the defining diameter of 'large wood', most previous studies (Table 1) have used larger spatial resolutions to detect wood accumulations. Indeed, Taminga et al. (2015) tested their ability to manually identify wood by down-sampling their native 0.05-m

pixel imagery, reporting that only beyond 0.8-m were pixels too large to accurately identify wood accumulations. Here, we find (and later report) that a great majority of accumulations of wood in the Elwha River consist of more than one piece and that we can detect many individual large wood pieces.

For each orthoimage, we used the digitized channel margins and river thalweg locations previously developed by East et al. (2018) to define the study area. The maximum extent of these combined channel margins was used to clip the extent of all model outputs. For ground-truth purposes, we hand-digitized all wood visible at 1:300 scale in four out of the 28 orthoimages: the first and last surveys in the record for both the middle reach and the lower reach. The digitizing scale was chosen to mimic the scaling of the orthoimage tiles used to train the model. These wood polygons were rasterized at the orthophoto resolution to be consistent with our gridded DL products.

3.2. Deep Learning label creation

The 28 orthoimages were split into 1024 x 1024 pixel tiles (each representing 128 x 128 m), with 50% overlap. This resulted in 45,878 image tiles for the middle reach, and 34,944 for the lower reach, or 80,822 image tiles in total. Labels were made using a Human-in-the-Loop image-labeling program, “Doodler” (Buscombe et al., 2021). Doodler facilitates relatively labor-free dense multiclass labeling of natural imagery, enabling relatively rapid training dataset creation for the four training classes (Figure 3 and Supplemental Figure S3). Data labeling was iterative; several versions of the DL model were constructed over the course of two years, each time adding new labeled data and retraining/evaluating the model output. This process only stopped when the model was deemed to be sufficiently accurate for the task. Ultimately, this decision was made using a comparative analysis of model-derived and ground-truth wood areas, outlined below in section 3.4. The final training dataset consisted of 4,382 images and corresponding labels, each 1024 x 1024 pixels and representing just over 5% of the total data set. The training data were sampled approximately equally in time and in space among both reaches. All training and validation samples purposefully included all four

label classes (vegetation, water, sediment, and large wood), to avoid model training and evaluation problems associated with class imbalance (Buscombe and Goldstein, 2022). Class imbalance is the term given to a situation where the model sees many more examples from certain classes compared to others, which leads it to be over-confident about the majority class(es), and a problem that could have easily manifested in this study if image tiles were drawn at random. Instead, imagery was selected based on a two-stage process. The first stage ensured that candidate image tiles were drawn from a stratified random sample based on both equal sampling of time, and from all locations within the river corridor of both reaches. The second stage relied on the image annotator (primarily this manuscript's first author) to only label candidate imagery that upon brief visual inspection was determined to contain all four classes.

3.3. Deep Learning model training

We use the software package Segmentation Gym (Buscombe and Goldstein, 2022) to fine-tune a Segformer (Xie et al., 2021) DL model architecture for the task of semantic image segmentation. We take the instance (i.e., model architecture and trained weights) of the model of Xie et al. (2021), itself fine-tuned on the ADE20k dataset (Zhou et al., 2019) consisting of 512 x 512-pixel tiles and fine-tune it on our 1024 x 1024-pixel training data. This is an example of the benefits of so-called transfer learning (Zhang et al., 2022), where one model instance trained for a specific task (i.e., to recognize classes in the ADE20k dataset) is fine-tuned to meet the needs of a similar task (i.e., trained under supervision to recognize the classes of a different labeled image dataset). This is achieved by initializing the model with the weights of the old dataset, modifying the last layer of the model to specify the new target classes and then iteratively adjusting the weights of the model during training using the new dataset.

Segmentation Gym allows for full reproducibility in the process of training or fine-tuning and evaluating image segmentation models, as well as hyperparameter experimentation for the purposes of optimizing trained models, as described by Buscombe and Goldstein (2022). The SegFormer model architecture was used, which

includes a hierarchical Transformer architecture, called “Mix Transformer”, as an encoder that extracts features from imagery, and a decoder for segmentation that uses those features for class probability estimation. The Mix Transformer is built around the concept of self-attention (Vaswani et al., 2017), and specifically a spatial self-attention layer. At a high level, a self-attention layer works in a similar way to convolution layers, highlights the salient spatial areas of the imagery important for feature extraction and prediction, while suppressing the irrelevant image portions for more contextualized predictions. It yields state-of-the-art performance while being more efficient to train than existing models (Rußwurm and Körner, 2020; Khan et al., 2022). Previous attempts to train a Residual UNet architecture, that used by Buscombe and Goldstein (2022), from scratch on this training dataset, resulted in a significantly worse performance and model convergence times, which is why the SegFormer model was ultimately used. The two model architectures crucially differ in terms of number of parameters, with the Residual Unet model having far fewer parameters because it consists only of convolutional layers, rather than fully connected layers (otherwise known as a perceptron in the parlance of traditional artificial neural networks) which have more parameters (Buscombe and Goldstein, 2022). The SegFormer model uses a fully connected decoder layer for classification, which we partially attribute to its success; it has far more parameters with which to make predictions, and the fully connected nature of the classifying layer means it can resolve classes at finer resolution. It is also possible that the basic image feature extraction process that the SegFormer uses to make predictions, the so-called self-attention layer (e.g., Aleissae et al., 2023), is more effective than the convolution layers used by the Residual UNet for the same task, which is consistent with earlier model comparison work using remotely sensed data such as Rußwurm and Körner (2020) but would require further investigation to verify. Unfortunately, so-called DL ‘model explainability’ and model calibration methods for image segmentation, which allow independent assessment of factors such as what image regions are important, and how under- or over-confident a model’s predictions may be for each class, are still in their infancy (e.g., Linardatos et al., 2020).

The segmentation example provided in Buscombe and Goldstein (2022) was an example of a model designed to generalize well to locations outside of the training

dataset; it, therefore, had relatively large numbers of samples for validation and relatively few for training. The reasoning for the abundance of validation data was to simulate a test set with a wider range of environmental conditions. In contrast, the concern with the present application is being able to segment imagery within a relatively small range of locations and times. While there is still considerable variability among sites and times owing to changing light, river color due to suspended sediment, river stage, and vegetation leaf conditions, we reasoned that model building would be improved with more emphasis on training samples than validation samples. In this manner, the model saw more of the data it will later, in effect, extrapolate over; thus, a smaller validation set is dictated. Therefore, the labeled images were further split into 3068 training and 1314 validation images. Unlike Buscombe and Goldstein (2022), which used augmented sets of training and validation imagery, here we used augmentation only on the training imagery to give the model more training data to generalize to, and to avoid the possibility of mis-representing validation statistics by using transformed validation imagery. Augmentation consisted of random horizontal and vertical flips of imagery, as well as shifts in horizontal pixels of up to 10% of the image width, and rotations up to five degrees. The final training data set consisted of 18,408 training and 1,314 validation samples.

We did not know in advance if wood detection would be better by a model trained only for that task, i.e., trained using binary labels of wood/other, or alternatively by a model trained to estimate all four classes (water, sediment, wood, vegetation) simultaneously. We therefore trained a multiclass and a set of binary models to compare approaches, evaluated in terms of the ability to detect wood. One consideration is to what extent the model needs a given class in the greater context of the adjacent classes; if this need is greater, we expected the multiclass model to perform better. Several model instances were trained for each case by varying the learning rate (Figure 4 and Supplemental Figure S4). All models converged well. In general, we found that a low learning rate ($1e^{-8}$) was optimal. The only hyperparameter we varied was the learning rate. Models were trained using a minibatch size of 12, the maximum size we could distribute across two RTX 3090 Graphics Processing Units, and model training typically

converged within 10-20 epochs (Figure 4; one training epoch typically completed within 30 minutes).

We used multiple standard metrics to evaluate segmentation model outputs, namely overall accuracy, frequency weighted Intersection over Union (fwIOU; Long et al., 2015), mean Intersection over Union (mIOU; Buscombe and Goldstein, 2022), Matthews Correlation Coefficient (MCC; Matthews, 1975), Precision, Recall, and F1 score, which are all standard evaluation metrics in applications of Machine Learning models (Maxwell et al., 2021). These quantify different aspects of the correspondence between observed and estimated wood pixels, leading Buscombe and Goldstein (2022) to conclude that keeping track of multiple metrics is optimal for final accuracy assessment as well as operational concerns such as monitoring model performance upon successive retraining with new data.

For comparisons among several models able to predict the same class, we found MCC better reflected the error rate visible to the eye than either mIOU and fwIOU, possibly because to yield a high MCC, relatively high scores are required for true negatives and positives, potentially ameliorating the problem of high recall (note that a model producing no false negatives has a recall of 1.0 masking low precision (a model producing no false positives has a precision of 1.0), and vice-versa, in F1 scores, which are the harmonic means of recall and precision scores (Maxwell et al., 2021). We also noted qualitative improvements in the visual assessment of outputs of the four-class model, which, while similar overall, tended to suffer from less egregious errors compared to the binary models (whose model training curves are shown in Supplemental Figure S4). We therefore chose to use the four-class model (Figure 4), which has a high MCC (Table 2), and is also more practical to apply than three separate models. From there, we evaluated Recall, Precision and F1 score to compare the per-class error rate for the four-class model (Table 3). However, we observe that these scores are heavily class-imbalanced (i.e., biased low for relatively rare wood and sediment, and high for relatively common water and vegetation classes). Ultimately, because this model is constructed as a measurement tool for quantifying wood primarily, and the major focus of this contribution is to use the model outputs to better understand the

dynamics of wood in the system, the accuracy of the model must also be understood principally in terms of the ability to estimate wood load over every spatial scale, and for the entire time period captured by the data. Therefore, the comparison of model-derived and ground-truth wood areas, outlined below in section 3.4, is the most important model evaluation metric.

To demonstrate change in model performance as a function of number of label images, separate four-class models were trained on datasets based on 999, 1999, 2999, and 4382 images, all with the same hyperparameters. Minimum validation loss, which is the final validation loss of the trained model on completion of training, decreased with the number of images (Figure 4), and all validation metrics increase (Figure 4). The MCC is overall the most sensitive metric for measuring model improvement over time with more available training data. Validation scores increase linearly with number of images (dashed lines in Figure 4c), whereas training scores plateau after 1999 labeled images. Training scores remain higher than validation scores, a reflection of our decision to provide the model with relatively more training data than validation data. Given much more model training experiments involving different proportional splits between training and validation subsets, the optimal split between training and validation would in theory manifest as more equal train and validation MCC scores (or other metrics). However, the disadvantage of these metrics is the difficulty in developing an intuition for their distribution of values, which vary from task to task, and do not scale with convenient variables such as dataset size, or number of classes. In practice, these metrics track relative progress of the model undergoing training under different scenarios (for example different/more data, or different hyperparameter choices) and are useful for determining when to stop labeling data, which is of enormous practical importance since labeling is time-consuming. However, ultimately, as we note above, in this study we conducted an independent assessment of absolute model skill with respect to large wood detection.

3.4. Image segmentation: Model implementation

To segment the 28 orthoimages into the classes, each image was split into 1024 x 1024-pixel tiles, with 50% overlap. The best four-class segmentation model was applied to each image tile, then outputs in the overlap regions were averaged to provide a more stable estimate of class-probabilities than would have been achieved without overlap. Overlapping tiles were then recombined into the full-size (i.e., the same extent and size as the original imagery) orthoimagery mosaiced into several raster bands, one per class. Averaging in regions of overlap was achieved using Lanczos resampling (e.g., Duchon, 1979) to enhance local contrast of classes at their mutual boundaries. These per-pixel, spatially averaged class probability maps were written to GeoTIFF format files, and the files representing per-class probabilities. The argmax (argument maximum) operator, which is used to return the maximum probability in the stack of four raster bands, was applied over each pixel to determine the class estimate for that pixel as the one with the maximum probability (Supplemental Figure S5). We used all four-class predictions for data visualization and transition analyses (please see section 3.7. below), and the wood and sediment predictions for further analysis of the evolving spatio-temporal distributions of wood and sediment.

3.5. Model wood detection evaluation

As noted in Section 3.1 above, all visible large wood in the active channel was digitized in four orthoimages of the study area. These spatially comprehensive data were rasterized and used as ground truth to evaluate the ability for the DL models to detect large wood. Observed and estimated reach-scale wood abundances were observed to be within 15% (Figure 5a), with 14.09% and 14.32% error, for middle and lower reaches respectively, in 2012-04-07, and 13.28% and 14.6% for 2017-09-22. Similarly, the observed and estimated size-distributions of wood areas compare closely (Figure 5b and c), which suggests that the model is good at detecting both smaller and larger wood pieces and piles without any systematic bias. Cumulative abundances as a function of distance downstream in the middle and lower reaches also track well, showing that error is accumulated throughout both reaches at a quasi-constant rate of about 15% (Figure 5d and e). We therefore recommend using this nominal, slightly conservative, error rate

of 15% for the entire spatio-temporal extent of the data, which reflects a zero bias in both space and time. We argue that this is a significant achievement considering the large variability in brightness and contrast owing to changing light and weather, and a very large variability in vegetation state and color over time, water color, amount of turbulent water, and other factors that vary considerably through the imagery time-series (see Supplemental Figure S2).

3.6. Metrics describing the magnitude, covariance, persistence of large wood

Large-wood magnitude is commonly reported as volume of wood per unit area and referred to as wood load (Van der Nat et al. 2003). However, measuring the thickness and porosity of wood accumulations still represents a major challenge (Campbell et al., 2019; Spreitzer et al., 2019; Livers et al., 2020). These quantities likely vary considerably in space and time due to natural variations resulting from transport and deposition processes, wood attrition and recruitment, and the diversity of the assemblage tree species and sizes (Campbell et al., 2019). Prior field-based and on-screen measurements of wood in the Elwha system (Leung et al., 2019; DeMott, 2021) strongly suggest a significant variation in wood thickness. Leung et al. (2019) estimated a 15% error rate in the specification of any dimensions of a large wood deposit, and such an error would quickly accumulate over an entire reach, especially when combined with an accumulation of a 15% error rate for wood detection. Neither the Leung et al. (2019) nor the DeMott (2021) studies examined porosity of large wood deposits. The thickness and porosity of a given wood accumulation cannot be derived with confidence from imagery alone, even though previous remote sensing studies have made these estimates using sphere approximations (e.g., Ulloa et al. 2015; Sanhueza et al., 2019) or using estimates of wood height (or thickness) and porosity (e.g., Livers et al., 2020; Livers and Wohl, 2021; Boivin et al., 2015). Finally, we note that DeMott (2021), who reported their Elwha wood measurements as areas for possibly the same reasons as listed above and were nonetheless able to draw convincing conclusions from their data regarding wood distributions in the middle reach. Ours is an evolution of the workflow demonstrated by DeMott (2021).

Here, our measurements of wood area were derived exclusively from enumerating wood pixels in the segmented imagery. Because we lack reliable information on the height and porosity of wood accumulations in the study area, we chose not to adopt a common approach of converting area to volume. Additionally, we have no basis to assume that average wood thickness or porosity values from other studies would apply to our study area, especially owing to the strong – and potentially unique – time-dependent patterns in large wood caused by the Elwha River dam removals, and a lack of applicable knowledge of typical porosity and height values for wood accumulations of specific tree species (cf. Campbell et al., 2019). For some applications, areas have been normalized by subregion of the active channel areas to provide the proportion of the active channel covered by large wood or sediment (m^2/m^2 , or %). This normalization was done within subregions by subdividing the active channel into 52 units in the longer middle reach and 44 units in the shorter lower reach that averaged 200 m in the along-channel direction. These regions were delineated in a GIS using the digitized channel margins previously developed by East et al. (2018). These measurements of large wood areas were obtained during low flow in the river (Figure S6, Figure S7a,b,c), and there was no significant correlation between estimated wood areas and flow ($r^2 = 0.11$ and 0.23 for lower reach and middle reach, respectively; Figure S7d), indicating that fluctuations in river stage did not significantly influence our results and interpretations. The discharge at time of the orthophoto surveys was always below the 25th percentile of the 5-year daily record for the study period (Figure S7c).

The geospatial data generated from the DL analyses were also combined into a datacube to assess multiple factors of the data at different spatial and temporal scales (cf. Hsu et al., 2015). These analyses include large wood distributions, magnitudes, and persistence, but they also include relationships between the deposition of large wood and sediment and the mode of wood storage. Wood persistence at a pixel level was quantified as the cumulative sum of wood for a given pixel location. When mapped, these cumulative sums provide an easily interpretable visual assessment. Wood-sediment relationships were assessed with the covariance between the respective time-series of wood and sediment magnitudes. Covariance was quantified using both wood-to-sediment and normalized wood-to-sediment ratios, as well as correlation analyses.

672 Normalization involved dividing the wood and sediment areas by their respective sums,
673 then computing the ratio of the two normalized quantities.

674 3.7. *Transition analysis and Markov chain model*

675 Our measurements consisted of full coverage (i.e., no data gaps) measurements at
676 the pixel scale, as well as a class ontology whereby a pixel cannot be more than one class
677 at a time, and there is no unknown or ambiguously defined class. As such, each 14-point
678 time-series representing the time-history of 13 discrete state-transitions at every
679 location in the river corridor are amenable to analyses of transition. Transition
680 probability matrices are a common tool to describe the likelihood of discrete state
681 changes in environmental systems (e.g., Davis and Sampson, 1986; Fieberg and Ellner,
682 2001; Takada et al., 2010). Here a state change is from one class to another e.g., water to
683 sediment or vice-versa would imply a reversible state change, i.e., sediment
684 deposition/erosion. A self-transition, such as wood to wood, is a direct measure of
685 persistence. The frequencies are compiled for each transition from each class to every
686 other class. Because there are four classes, a 4 x 4 transition probability matrix was
687 made to represent the empirical likelihoods of 16 possible processes encoded at every
688 pixel location and every timestep. Here they are used to examine the likelihood of
689 individually relevant changes in state, for example from sediment to wood which
690 indicates wood deposition, or from sediment to vegetation which indicates vegetation
691 growth and sediment stabilization. Additionally, they are used to show downstream
692 trends in transitions, and reach-wide differences in average behavior (a ‘fingerprint’ of
693 the system at equilibrium).

694 The timing of the present study is such that it covers the tail-end of the period of
695 dam removal, which began in 2011, and the subsequent rapid evolution of the system
696 toward a new equilibrium state (DeMott, 2021). Transitions in the Elwha system are
697 amenable to stochastic analyses of our dataset, such as Markov chain analyses, that may
698 capture this transition to a new equilibrium state for each class and across each reach
699 and sub-reach. The steady state probability distribution, computed using a Markov
700 chain model constructed from a transition probability matrix, describes the system
701 which has converged to a point where the transition probability distributions will no

longer change, i.e., equilibrium. In other words, the time-invariant probabilities of a change to or from vegetation, water, sediment, and wood. This distribution does not depend on the absolute amount of any class. The changes encapsulated by our data conform to the Ergodic Theorem for Markov Chains (Davis and Sampson, 1986), where states are irreducible, recurrent, and aperiodic. In this scope we define the four classes to be states. States are irreducible when it is possible to get to any state from any state. This is true for the Elwha River because any state in a specific location can be replaced in that location over time through physical (e.g., wood deposition or removal) or biological (e.g., vegetation growth or removal) processes. We also reliably assume that the system is statistically recurrent, that is, there is a non-zero probability that any state can return to its starting state. Finally, the system is aperiodic, which is the ability of the system to return to a given state over a single time step.

A time-homogenous Markov chain is defined in terms of the probability distribution of state transitions. Specifically, the probability distribution of states at time step $t+k$ is given in terms of its distribution at time t by (Davis and Sampson, 1986),

$$\pi(t + k) = \pi(t)P^k, \quad (1)$$

where all information about the evolution of the probability distribution of m states is contained in the $m \times m$ transition probability matrix, P , a row-stochastic matrix where all rows sum to 1. The probability distribution π is the steady-state distribution. If Markov Chain theory holds, the k -step transition matrix should be approximately P^k . This presents an opportunity to use the Markov chain to test for the hypothesis of equilibrium, by comparing individual $P(t+k)$ compiled from the data, with P^k determined from (1). The steady-state distribution satisfies

$$\pi = \pi P, \quad (2)$$

where π is computed as the left eigenvector of P (the eigenvector associated with an eigenvalue of 1).

We compile P for each sub-reach of the middle and lower river reaches, using the same sub-reach system as used for computed normalized wood and sediment area

measurements. To compare the average tendencies of class transitions in both reaches, we computed the mean probability transition matrix by aggregating transition frequency matrices per sub-reach, before normalizing by reach-wide frequencies of each class to estimate the average P per reach. Transition data per sub-reach are also visualized as a function of downstream distance. Finally, we compute and compare π for each reach to examine and discuss any reach-wide differences in the steady state probability vector. Similar analyses were employed by Senter and Pasternack (2011) to examine the role of large wood in salmonid spawning habitat.

3.8. Supporting data

The Lower Elwha Klallam Tribe has overseen the construction of Engineered Log Jams (ELJs) in the lower reach and have collected data on several channel attributes in and around the ELJs. The Lower Elwha Klallam Tribe data used here includes construction date and location of ELJs, and maximum pool depths and bar grain size distributions in the vicinity of ELJs, which were obtained at various times between 2001 and 2021. Grain-size distributions were determined using a standard Wolman count based on randomly selected surface particles ($n=100$). Lastly, river discharge and sediment load measurements during the period come from Ritchie et al., (2018b). These data quantify fluxes of water, suspended sediment, and bedload sediment at U.S. Geological Survey gaging station 12046260, which is downstream of the two former dams (~1.5-km downstream of Elwha Dam site).

4. Results

4.1. Large wood and sediment abundance

The segmented imagery provides detailed time histories of the evolution of the Elwha River channel margin during and after dam removal, and example DL results from several channel reaches are provided in Figure 6 highlighting only the net change

over the 5.5-year study period. In general, the imagery and its DL segmentation show that all channel sediment bars expanded in size and position over the orthoimagery record, and that the amount and distribution of large wood also changed markedly on every bar, often several times during the 5.5-year time-series (Figure 6). Common patterns of changes to the fluvial landscape include: (i) wood deposition concentrated at the upstream apex of bars constraining flow, (ii) river channel position changes associated with appearance of new wood deposition, and (iii) a decrease in the areal extent of vegetation and water, which are consistent with the observations of Ritchie et al. (2018a) and East et al. (2018). There was no evidence of any channel-spanning log jams, nor observable hydraulic backwater effects (cf. Livers and Wohl, 2021) associated with wood accumulations and no evidence of wood rafts that completely blocked the active channel (Boivin et al., 2015). Only a negligible (estimated <1%) amount of wood was detected in the channel, presumably undergoing active transport at the time of survey (and could possibly be further isolated and studied by finding wood pixels surrounded by water pixels).

The average spatial patterns of large wood and sediment provide an overview of where accumulations were found, and how this spatial distribution changed in time (Figure 7). In the middle reach, large wood was concentrated near the center of the reach, where it reached total areal maximums at about 16-rkm (Figure 7a). In the lower reach, total large wood abundance tended to increase in the downstream direction resulting in areal maximums in the wider, lower-gradient distal end near the river outlet (0- to 5-rkm; Figure 7a). The total area of both large wood and sediment as a function of distance along the river were highly correlated (Table 4) because almost all large wood was deposited on bars (Figure 7b), rather than in the relatively short canyon sections where wood is occasionally found in riffles. The linear correlation (r) of these time-averaged wood and sediment areal values across the 96 subregions presented in Figure 7 was 0.88 and 0.92 in the middle and lower reaches, respectively (Table 4). Correlations between the time-series of reach-wide wood and sediment areas are also high ($r=0.86$ and 0.77 for middle and lower reaches, respectively; Table 4), but slightly lower than correlations over downstream distance. Finally, time-series of wood in the two reaches also display a high correlation ($r=0.78$), as do time-series of sediment ($r=0.89$),

suggesting that both reaches responded similarly to the same upstream sediment and flow boundary condition without significant time lags. These results suggest strong colocation relations between the sediment bars and deposits of large wood, which we will explore more fully below.

Reach-wide totals of areal values of large wood and sediment changed significantly over time (Figure 8). Whereas wood magnitudes plateaued in both reaches in 2014 or 2015, the sediment areas in these reaches continued to increase through at least the winter of 2016/17 (Figure 8a, b). Large wood areas increased from around 15,000 m² in the middle reach and 20,000 m² in the lower reach to approximately 35,000 m² in both reaches, most of which occurred between April 2012 and the winter of 2014 (Figure 8b). In contrast, areas of deposited sediment increased from 100,000 m² in the middle reach and nearly 200,000 m² in the lower reach, to ~250,000 m² in both reaches by 2014 and to more than 300,000 m² in both reaches by 2017 (Figure 8b). These up to four-fold increases in the areal extents of both large wood and sediment greatly exceed measurement uncertainties of ~15%. Over the study period, the middle reach saw largest net proportional increases in wood and sediment, with 2.3-times more wood, and 3.5-times more sediment at the end of the study period than at the start. The equivalent net increases for the lower reach were a 1.75-fold increase in wood and 1.5-fold increase in sediment. This suggests that, while the time-series of both quantities in both reaches are similar in magnitude, the lower reach saw greater net increases in wood, whereas the middle reach saw greater net increases in sediment.

As noted above, there was strong correlation between the areal extents of large wood and sediment, although further examination of these data suggests that the total ratio of large wood to sediment ratio declined over time (Figure 8c, d). For example, the ratio of wood to sediment peaked at 0.175 in the middle reach in July 2012 and declined to less than 0.1 by September 2017 (Figure 8c, d). In the lower reach, the wood-to-sediment ratios declined from 0.16 to 0.10 in the same interval of time. Normalized wood sediment-to-ratios show a very similar pattern, although these data show that the overall normalized ratio of wood-to-sediment in the middle reach was about 50% greater than the values measured in the lower reach (Figure 8b, d). Overall, this suggests

that total areal extent of the exposed sediment on the channel bars continued to increase after the area of wood deposition had plateaued.

A more detailed examination of the spatial and temporal context of large wood and sediment abundance is provided in the matrices shown in Figure 9, where the data are presented as raw areal values (upper figures) and normalized areas by including the local channel margin areas (lower figures). Normalizing these areal measurements reveals that sediment commonly dominated the channel margin following the influx of 2012-2013, when it regularly exceeded 50% of this area (Figure 9d). Large wood typically covered ~5% of the margin area, although the largest wood accumulations could cover up to 25% of the available area at the local sub-reach scale (Figure 9c). The temporal fluctuations in large wood and sediment reveal several localized deposition events in both the middle and lower reaches, for which wood areas could increase 2 to 10-fold during the events (Figure 9; events highlighted with arrows). For example, several localized increases in wood abundance were observed in the middle reach during 2013 and 2014, and another large wood deposition event occurred at 17- and 18.5-rkm in 2016 (Figure 9a,c). The event-based deposition in sediment is similar in timing and location to the event-based deposition of large wood, suggesting that these two phenomena are strongly coincidental (Figure 9).

4.2. Persistence and size distributions of large wood

A spatial method to visualize persistence is to enumerate each grid cell with the proportion of time that location was occupied by wood. In this context, the ever-persistent wood piles will have a value of 1 and may be long-standing natural logjams or ELJs, whereas values of 0.5 would suggest that the wood had been present half of the transpired time. Examination of the temporal patterns of wood in the Elwha River channel margin revealed several patterns of persistence, exemplified qualitatively in Figure 10 (and Supplemental Figure S8). Some channel reaches exhibited more spatially dispersed accumulations of wood, for which wood pieces were observed to be individual pieces of accumulations of only several logs (Figure 10 and Figure S8).

An examination of the evolution of size-distributions of individual wood accumulations could reveal important processes responsible for the dispersal of wood (e.g., Kraft et al. 2011). For example, an increase in the size of wood deposits may suggest a concentration of wood due to trapping of transported wood pieces, perhaps in regions of flow convergence. Conversely, a decrease in wood deposit size may indicate dispersal of wood, perhaps due to local sediment erosion or distribution of wood pieces in reaches undergoing fluctuating or receding flows (Wohl et al., 2019a; Wohl et al., 2023). These quantities could be examined for each individual wood accumulation, but instead we focus on the reach-wide trends of wood size distributions to examine the general tendencies in each reach for comparative purposes. The time-series of reach-averaged wood size-distributions show similar trends, with the coarse tail of the wood size-distributions relatively depleted in 2012, then enriched by 2014, then relatively depleted again thereafter (Figure 11a, b). Wood size distributions are exponentially distributed in both reaches with an exponential fit explaining all but the smallest wood pieces and piles (Figure 11c). This observation may be useful for developing models for large wood size distributions in similar rivers. Mean wood accumulation area varied between 7- and 24-m² over the study period (Figure 11d), increasing rapidly until its peak in summer 2013 and decreasing slowly thereafter. This general pattern was stronger in the middle reach and weaker in the lower reach.

4.4. Engineered Log Jams

The dynamics of wood in the vicinity of known ELJs in the lower reach was investigated to determine whether the DL techniques could adequately map and characterize these features and to assess the time history of the 55 structures during the channel adjustments such as avulsions and bar building that resulted from dam removal. Several ELJs constructed prior to dam removal were already gone (or not detectable because of obscuration by vegetation) by the start of the present study in April 2012 (Figure 12a, c). The amount of large wood in the reach that was deposited outside of the ELJs was far greater than the wood deposited in the vicinity of ELJs (Figure 12b), especially in the Hunt Road channel or western anabranch (a major

anabranch ~1-km long in the lower reach that became the mainstem channel after a 2007 flood; Draut et al., 2011). However, extra-large wood deposition did occur at the three ELJs from the pre-dam era that persisted until September 2017 (Figure 12c). The number of detectable ELJs within the river channel margin declined over time from a maximum of nine (Figure 12d). Active ELJs are defined as those with wood accumulations within a 5m radius at time of aerial survey. We also note that several of the ELJs may have been obscured under canopy, and their disaggregation was not captured by the frequency of our imagery, so we cannot be certain about what happened to many of them. However, it is clear that they are playing a less important role in river morphodynamics, because many have demonstrably gone, whereas those obscured by canopy may now be disconnected from the river under most flows.

4.5. Channel characteristics in the vicinity of wood accumulations

Time-series measurements of numerous alluvial bars in the lower reach made by the Lower Elwha Klallam Tribe reveal a dramatic overall fining of alluvial bars until 2014, then a subsequent coarsening of those bars until 2021 (Figure 13b). This is consistent with hundreds of field measurements of surface grain size made in three sub-reaches of the lower reach by East et al. (2018; their Figure 4), where sediment fined greatly in 2011–2014 compared to the dammed condition (measurements from 2006 to 2011). Grain size coarsened gradually after 2015 but remained finer than in its dammed state by 2017 (East et al., 2018). This indicated that the peak of the sediment pulse from upstream reservoirs has passed through this location, and that the river had returned to a supply-limited state after having been transport-limited during the peak of the sediment pulse (East et al., 2015, 2018). However, mean wood rack volume increased steadily until 2018 (Figure 13c), as did pool depth until 2021 (Figure 13c). We also observed a decline in mean pool depth in 2014/2015 coincident with peak sediment transport, and then a deepening trend to 2021 (Figure 13d).

4.6. Transition analysis

The full- and sub-reach scale transition probability matrices and steady state probability vectors were computed for both reaches following the methods outlined in section 3.7. A comparative analysis of reach-wide transitions is presented in Figure 14. Each matrix can be viewed as encapsulating the likelihoods of 16 unique processes associated with channel change (Figure 14c). In addition, the differential between the respective probabilities in middle- and lower-reach matrices (Figure 14d) provides a ready means with which to contrast the two reaches in terms of the relative importance of each process in each reach. The reach-averaged transition matrix, P , for the middle reach (Figure 14a) and lower reach (Figure 14b) display significant differences in the probabilities of transition between several classes. Middle reach wood is globally more persistent, with a self-transition probability of 0.24 compared to 0.18 in the lower reach. Lower reach sediment is also less persistent, with a value of 0.43 compared to 0.51 in the middle reach. Among other noteworthy differences are those for vegetation to wood transitions, many of which are due to vegetation removal and subsequent wood deposition in that location, processes which are twice as likely in the middle reach (0.24) compared to the lower reach (0.12).

Transition probabilities for each of the 16 processes at sub-reach scale are summarized in Supplemental Figure S9, providing a comprehensive picture of the occurrences and rates of individual processes as a function of downstream distance. Persistence in wood can be highly localized in both reaches (Figure S9a). Wood persistence is generally associated with high persistence of sediment at sub-reach scales, especially in the middle reach. Persistent wood in the lower reach is more spatially variable than compared to the middle reach having a spatial variability value of 1.0 in the lower reach and 0.86 in the spatially less variable middle reach (Figure 14e, f).

The prevalence of processes that add or remove wood also vary considerably over increasing distance downstream (Figure S9c, d). However, it is apparent that wood deposition rates in locations previously occupied by water is significantly higher in the lower reach (Figure S9c), as are wood removal rates. In short, there is a greater rate of transition from water to wood and vice-versa in the lower reach compared to the middle reach. Conversely, the rate of transition from vegetation to wood, and vice-versa, are

significantly more common in sub-reaches of the middle reach, compared to equivalent data in the lower reach. These data suggest that the coincident enlargement of bars and wood deposition in the middle reach occurred predominantly in portions of the floodplain that were previously vegetated, whereas wood/sediment deposition in the lower reach predominantly occurred in areas previously occupied by river channel. These are the reasons why the inter-reach probability divergence values for vegetation to wood, and vegetation to water, are large (Figure 14d).

The steady-state vectors for the middle and lower reaches were computed following section 3.7. These vectors represent the time-invariant probabilities of states in a system assumed to be in steady state (equilibrium), or the long-term likelihood of transitioning to/from a class. In the middle reach, $\pi = 0.57, 0.048, 0.22, 0.17$ for vegetation, water, sediment, and wood, respectively. The same vector for the lower reach is $\pi = 0.65, 0.07, 0.07, 0.21$. The values indicate a degree of similarity in tendencies for transitions to vegetation, water, and wood. The most significant finding is that 17 and 21% of transitions in the middle and lower reaches respectively involve large wood, underlining its importance in the morphodynamics of this river. The most notable difference is the comparatively low value for lower reach sediment (0.07) compared to that in the middle reach (0.22). These steady state probability vectors may be viewed as a fingerprint of the system at hypothetical equilibrium. Neither reach is a closed system with respect to large wood and sediment due to transport into and out of either reach, e.g., wood can be recruited from floodplain areas during channel migration and high flows, and wood in transport could also be deposited on the floodplain. With no indication that the lower reach was more out of equilibrium than the middle reach, the relatively low value for steady state sediment probability may reflect the fact that the lower reach saw greater net increases in wood compared to sediment (Figure 8) and that sediment in the lower reach is less persistent than in the middle reach (Figure 14). Like many of the complementary comparative analyses presented here between the two reaches, the respective reach-wide steady state probability vectors encapsulate both the similarities and differences of landcover distribution dynamics, which are summarized using all lines of evidence in section 5.2.

966

967 5. Discussion

968 The new machine-learning methods utilized in this study provide for an
969 unprecedented level of detail in examining the evolution of large wood in the Elwha
970 River corridor over both multiple years and tens of kilometers. Moreover, the unique
971 opportunity to study a river undergoing a planned major pulse of sediment and wood
972 during the world's largest dam removal to date provides a unique before-after
973 investigation of large-wood movement. Below, we explore some of the implications of
974 our data from the Elwha River for understanding morphodynamic evolution of wood
975 and sediment transport, a comparison of the responses of both reaches to their
976 respective dam removal processes, and finally a discussion of the broader implications
977 of our methods and findings for large wood management and research.

978

979 *5.1. Morphodynamics of Elwha River wood and sediment.*

980 Large wood deposition in the Elwha River floodplain increased significantly between the
981 start of dam removal (2011) and winter 2013, then plateaued (Figure 8b). This pulse in
982 wood occurred simultaneously in the lower reach and middle reach during the peak
983 sediment load and slightly after when geomorphic changes in the river were at their
984 maximum (in Spring 2013, as reported by East et al. (2018)), resulting in wide, shallow,
985 braided channels and meters of bed aggradation (East et al., 2018). By 2017 the river
986 had begun to incise into its new deposits (East et al., 2018) and appeared to be in
987 dynamic equilibrium with regard to wood storage and export. This is apparently
988 consistent with wood mobilization studies that report total storage of large wood may
989 remain nearly constant (Marcus et al., 2002; van der Nat et al., 2003; Wohl and Goode,
990 2008; Wohl and Scamardo, 2021; Wohl and Iskin, 2022). However, there was an
991 increase in sediment deposition beyond the wood plateau, up until 2016/17 (Figure 8d),
992 which suggests that sediment bars continued to expand in areal extent, helped at least in
993 part by an up to four-fold increase in wood deposition over a short period of time. This
994 observation is consistent with studies demonstrating sediment transport suppression

and sediment deposition through increased hydraulic roughness due to the introduction of large wood (e.g., Davidson and Eaton, 2013; Wohl, 2017). Large wood covered up to 25% of the available area at the local sub-reach scale, almost always accumulating on top of sediment that would be otherwise available to transport by flows. Further, the largest and most persistent wood accumulations in both reaches are strongly associated with larger braid plains where the river has multiple channels. This is consistent with the observation of Scott and Wohl (2018) in a Pacific Northwest river that wood jams are significantly more likely in deeper and multithreaded channels than shallow and single threaded channels. There is some evidence that a very small portion of wood undergoes at least temporary burial by sediment, from visual evidence of wood appearing partially buried. However, such instances are very rare, and model detection is variable for these partially buried pieces, so we attribute the contribution of such wood to total area as within the 15% reported error. The transition probabilities from wood to sediment constitute 13% and 12% of transitions to sediment in the middle and lower reaches, respectively. These numbers reflect both cases of true burial of wood by sediment, as well the transportation of wood away, exposing sediment.

Our measurements provide an integration of both logjams and individual wood pieces, such that we cannot separately report the abundances of each sub-population. However, our remote-sensing observations suggest that the size distributions for wood depositions varied considerably in time and space (Figure 11). The most notable finding was an initial increase in average wood deposit size coincidental with the initial sediment pulse, especially in the middle reach, which was followed by a transition toward smaller average deposits. These findings are broadly consistent with those of DeMott (2021) whose dataset consists of six aerial surveys over the middle reach. They note that the sum of logjam area values increased steadily until 2016/17. Additionally, they note that the number of individual logs, a quantity not measured here, spiked in summer 2013 and then fell back down to pre-dam levels. We observe the same initial increase in wood (Figure 9a) that is consistent with the delivery of a pulse of individual logs from the reservoir (Leung, 2019), as well as coinciding with the peak in sediment load (East et al. 2018).

Wood debris is known to enhance pool formation (e.g., Stewart et al., 2012; Livers and Wohl, 2021), and the fact that sediment bars continued to expand in areal extent until 2016/17, whereas large wood was deposited predominantly during the period up to winter 2013, may be due to roughness-induced sediment transport suppression such as discussed above. Alternatively, our observations may suggest that the supply of wood was extinguished faster than the supply of sediment. Using our measurements, which provide total wood storage but do not include rates on large wood import or export in either reach, this scenario is more difficult to evaluate, but is also plausible. A large but finite amount of wood would have accumulated in both reservoirs during the dam period and local production of wood would be a function of channel change rates, which slowed after peak rates in 2012-13 (East et al., 2018). In addition, dam removal allowed for renewed wood transport from the upper watershed to the middle and lower reaches, so the source of wood to the study area would be a combination of these three complex supplies. There was significantly more deposition of wood in previously vegetated areas of the middle reach than the lower reach (Figure 14), which provides evidence that channel enlargement may have promoted wood recruitment and deposition. The reservoir population of wood presumably dominated the large wood signal in the years during and following dam removal, suggesting pulses in phase with the erosion of wood-bearing deposits of sediment within each reservoir. Upper watershed wood supply is largely unknown. That noted, the residence time of wood undergoing active transport is unknown, promoting uncertainty in how long it would take a wood pulse to travel through the system. We see a very broad wood size distributions (Figure 11), implying a wide spectrum of large wood disaggregation and transport rates. We see a temporary increase in average wood accumulation size associated with the period of reach-wide wood enrichment (Figure 11d), perhaps suggesting a discrete period of wood concentration into larger jams, followed by disaggregation and dispersal. The peak in wood abundance coincides with the peak in average wood deposit size, perhaps partly due to recruitment of wood from floodplain areas that the river accessed by channel migration during peak geomorphic change rates in early 2013 (East et al., 2018). The four years after dam removal saw an approximate doubling of peak flow (from ~200 m/s³ in 2012 to ~400 m/s³ in 2016; Supplemental Figure S7a), which could have recruited wood to the channel through bank erosion. We also see a large amount of

persistence in wood, suggesting that many wood accumulations were persistent through a large range of flow and sediment conditions, but also a large degree of local variability in wood and sediment persistence (Figure S9a). Disentangling the sources of the wood may be possible with the higher frequency imagery that have been collected but have not been processed into orthoimagery (Figure S1). These data coupled with more detailed analyses may provide further clues to the sources, transport, and fate of wood in the Elwha River.

5.2. Comparison of wood and sediment dynamics in the middle and lower reaches.

It is instructive to synthesize the similarities and differences in response to dam removal in the two reaches to inform both dam removal processes in other comparable systems, as well as to inform habitat and restoration studies. In both reaches, all sediment bars underwent a net enlargement because of dam removal, and newly supplied wood almost always deposited on sediment bars rather than on floodplains, riffles, and steep channel margins. Magnitudes in time-series of reach-averaged wood and sediment totals were remarkably similar (Figure 8), despite differences in reach length, slope, sinuosity, and timing of dam removal. Additionally, the size-distributions of wood accumulations in both reaches showed similarity in exponential form, as well as broad similarities in reach-wide time-series of mean wood deposit size (Figure 11). Strong correlations were found between the time- and spatial-series of both wood and sediment in both reaches, as well as time-series of those quantities between reaches (Table 4). The values and time-series of ratios of sediment to wood in both reaches were similar. Finally, similarities existed in the rates of many processes revealed by an analysis of transition (Figure 14), for example sediment erosion and deposition rates, vegetation burial and wood deposition. We attribute these similarities to the magnitude of shared upstream flow and sediment boundary condition during peak sediment load which saw similar geomorphic processes simultaneously occur in both reaches (East et al., 2018), aided by similar distributions of sediment grain size in both reaches (Draut and Ritchie, 2015) and relatively steep slopes.

1086 However, noteworthy differences were also revealed in the response of the two
1087 reaches to dam removal. Net increases of wood over the study period were slightly
1088 greater in the lower reach, whereas net increases in sediment were greater in the middle
1089 reach (Figure 8). A potential reason for this is the comparatively steeper slope of the
1090 river in the middle reach, implying greater stream power that is expected to increase
1091 throughput of sediment and wood, enhance hydraulic roughness, and suppress bar
1092 building compared to the less steep lower reach. The middle reach saw a greater
1093 increase in average wood deposit size (Figure 11), which may have further locally
1094 increased channel roughness and promoted sediment deposition. While present, this
1095 signal in the lower reach was more muted, despite the existence of ELJs in this reach,
1096 perhaps due to less flow convergence and accumulation in areas of large wood
1097 accumulations, or less dispersed wood available to source wood pile growth during the
1098 time of maximum flows.

1099 Wood in the middle reach was more persistent overall (Figure 14), but persistence
1100 was also highly localized (Supplemental Figure S9). Sediment in the lower reach was
1101 less persistent, and persistent wood was spatially more variable in that reach (Figure 14,
1102 15). We attribute the relative persistence of wood in the middle reach to many of the
1103 initial deposits being formed at relatively high elevation, under elevated flows and
1104 sediment loads of the initial period up to winter 2013 when most of the new wood from
1105 upstream was deposited. A significant amount of this wood is in the middle portion of
1106 the middle reach, whereas most large wood in the lower reach is closer to the river outlet
1107 and its persistence lower due to more propensity for wood transport over the generally
1108 flatter, lower elevation bars with a larger range of stages for a given flow discharge
1109 range.

1110 1111 *5.3. Potential benefits of reach-wide pixel-wise measurements for large wood* 1112 *science, and future work.*

1113 Reflecting on 50 years of large wood research, Swanson et al. (2021) conclude that
1114 “Given the documented differences in large wood dynamics across different portions of

a river network and between networks in diverse geographic regions, our understanding of large wood in rivers would benefit greatly from coordinated, longterm field studies in multiple field sites that focus on a few key questions. This type of research programme could foster direct comparisons among sites.” Automated measurements of large wood, such as presented here, could arguably make that goal more likely. Our approach to image segmentation using deep learning could likely scale to other similar river systems and similar imagery. The prospect of developing accessible DL-based image segmentation models that may transfer outside of their training contexts, i.e., a specific location, image source, or thematic context, is a potentially significant development for riverine wood science. We therefore intend for our open-source image segmentation software (Buscombe and Goldstein, 2022) and publicly available models (Buscombe, 2023a) and datasets (Buscombe, 2023b) to stimulate further application and research into simultaneous large wood dynamics in several catchments, as well as to facilitate further research on the Elwha River. This methodology could be used to quantify all wood visible at the surface, especially useful in relatively inaccessible rivers and parts of river systems. Capturing wood over all spatial scales provides an opportunity to examine the role of spatial dispersal in wood transport, formation of jams, trapping efficiency (Wohl et al. 2018), as well as any morphodynamic role of wood that is variously dispersed, in the formation of bars and islands (Piégay 1993; Gurnell et al., 2019).

Because reach-wide pixel-wise measurements facilitate comparison of wood and sediment storage at any scale or location, routine monitoring of large wood at large scale could become an important component of characterizing the natural wood regime of a river. It could prove useful in planning and implementing the removal and construction of structures such as dams and fish passage and establishing the target wood regime for that river (Wohl et al., 2019a). Once monitoring protocols are established, we can continue to learn about the natural recruitment and dynamics of wood in situ, through refining techniques to detect incremental change, as well as adequately capturing catastrophic disturbances that are associated with abrupt increases in mobilized large wood, such as dam removal or landslides (Ruiz-Villanueva et al., 2014). This approach does not provide useful information on buried large wood, or wood obscured by canopy,

unless a high-frequency dataset is used that captures these processes as they happen, but each orthoimage can provide a near-complete synoptic survey of large wood. In the future, addressing any wood load bias due to seasonal or permanent canopy may be achieved using in situ measurements. The extent to which specific large wood dynamics can be elucidated from the time-series depends on the specifics of the undergoing changes and how that intersects with the timing or return interval of the imagery.

Our techniques and insights may be useful for studying the morphodynamics of wood-sediment interactions in other geomorphically complex channels. For example, Scott and Wohl (2018) note that residence times (persistence) remains a major knowledge gap in mountainous watersheds, and Wohl (2020) notes that more knowledge about deposition modes in individual systems would help local wood restoration strategies. Dispersed pieces are assumed to be more mobile, and the distribution among dispersed and concentrated pieces in the large wood population exchanged between active and stored subpopulations is, at present, largely unknown. The conversion rate between mobile and stored wood populations, assuming there is a sub-population of mobile wood that gets trapped by stored wood (Gurnell et al., 2000) is also not generally known or parameterized. However, techniques like those presented here could be used to identify and track individual pieces or accumulations of large wood, from a very high-resolution time-series of imagery, for example weekly or monthly resolution, and compute the magnitudes of the mobile and stored subpopulations. Finally, perhaps soon it will be possible to infer more large wood characteristics using automated methods, for example the complexity of individual pieces, or classifying wood by age (to infer wood recruitment and the effect of transport processes, among numerous potential uses).

The magnitude of wood stored is the balance between inputs and outputs. Closing a large wood budget for a river system (Benda and Sias 2003; Lucía et al. 2015; Comiti et al. 2016) requires accurate measurements of inputs (wood arriving from upstream and local areas), and outputs (in this case, wood exiting to sea) (Boivin et al., 2017). Comprehensive measurements of wood occurrence, i.e., for every pixel in an image dataset that covers the full spatio-temporal extent of interest, or ‘datacube’, facilitates a

different approach. It allows us to measure the storage term of the budget directly, which circumvents the rather more difficult task of measuring how much wood enters and leaves a system, then differencing those two quantities. In other words, while we do not have quantitative estimates of the amounts of wood entering the reaches from upstream, and how much wood exited to sea, we can potentially measure the changes in wood storage with high-frequency imagery.

Future work should use data such as those presented here to map the orientation and other potentially important characteristics of large wood deposits, to infer processes at reach scales. In a study conducted in the channel occupying the former reservoir and immediate downstream, Leung et al. (2019) concluded that the primary control on fluvial scour hole size is the presence or absence of roots, with scour pool volume increasing with root cross-sectional area. The shape and orientation of large wood deposits were found to have additional effects on scour pool depth, volume, and shape. In contrast, variability in flow discharge was found to be a limited control on scour pool size. While our measurements are limited by a lack of volumetric information about large wood deposits, further studies could potentially examine these conclusions further using our methods and data to measure the shape and orientation of wood accumulations, and it would also be possible to manually identify larger root wads. Preferential wood orientation occurs due to the transport of a large wood usually occurring with the rootwad pointing downstream, and upon deposition, the roots anchor the wood into an orientation parallel with flow (Abbe and Montgomery, 2003; Davidson and Eaton, 2015). Leung et al. (2019) confirmed this phenomenon in the Elwha River. Therefore, a time-series of wood orientation measurements at individual locations in the Elwha system may reveal the chronology of wood deposition with respect to flow fluctuations or might even be helpful in developing stage-discharge relations if the elevations of recently deposited wood can be mapped.

A spatially comprehensive approach would also allow for assessment of spatial-temporal factors such as the so-called “mode” of wood stored within the river corridor, which may be either dispersed (i.e., individual logs) or concentrated (i.e., logjams) (e.g., Kraft et al. 2011). The mode of stored wood may vary considerably over spatial scales

(Abbe and Montgomery 2003, Scott and Wohl 2018) and is related to recruitment and transportation processes (Wohl et al., 2019a; Wohl et al., 2023).

Deep learning models typically require large volumes of labeled training data; the only other known example of large wood detection using automated supervised deep learning methods, namely that of Sendrowski et al. (2023), used 10,053 hand labeled image tiles. However, once the model is trained on a larger number of examples, it is much more robust to noise than other available methods. Given the relative differences in study area (Table 1), our training data consists of between 5 and 25 labeled images per hectare depending on the amount of labeled imagery used for the model (Figure 5f), and that of Sendrowski et al. (2023) consists of an estimated 0.28 labeled images per hectare. It would be beneficial to combine labeled datasets and train new wood detection models under Transfer Learning scenarios where multiple sources of imagery are available, such as to construct time-series from disparate image sources and times.

In this study, our use of the argmax operator to determine most probable per-pixel class as the one with the largest probability was a standard approach in image segmentation (Buscombe and Goldstein, 2022). However, this approach does not provide an associated measure of relative uncertainty of each estimate, such as the post-hoc classification of an ‘uncertain’ or ‘unknown’ class where probabilities do not hit a threshold, which could be addressed in future work. Additionally, the use of coincident elevation data in future work would lead to a greater understanding of wood deposits in relation to the elevation of river level. Airborne light detection and ranging (lidar) or Structure-from-Motion (SfM) photogrammetry digital surface models (DSMs) would therefore be a valuable additional data source (e.g., Kasprak et al., 2012; Spreitzer et al., 2019; Sanhueza et al., 2022).

6. Conclusions

We developed new machine-learning methods to provide for fully automated planform measurements of large wood, sediment, vegetation, and water from a time-series of whole-river-corridor, decimeter-scale orthoimagery. These measurements

1235 provide for an unprecedented level of detail in the examine the evolution of large wood
1236 in the Elwha River corridor over multiple years and tens of kilometers, accurate to
1237 within 15%.

1238 Large wood deposition increased between the start of dam removal (2011) and
1239 winter 2013, then plateaued. Sediment bars continued to grow up until 2016/17, assisted
1240 by a partially static wood framework deposited predominantly during the period up to
1241 winter 2014/15. Large wood typically covered ~5% of the channel margin area, although
1242 the largest wood accumulations could cover up to 25% of the available area at the local
1243 sub-reach scale. Mean wood accumulation area varied more than three-fold over the
1244 study period, increasing rapidly until its peak in summer 2013 and decreasing slowly
1245 thereafter. A Markov chain analysis of transition revealed that transitions to/from wood
1246 account for up to 21% of all transitions, underlining the importance of wood in this
1247 dynamic system.

1248 While the time-series of both wood and sediment in both reaches are similar in
1249 magnitude, the lower reach saw greater net increases in wood, whereas the middle reach
1250 saw greater net increases in sediment. The total area of both large wood and sediment as
1251 a function of distance along the river were highly correlated. The same was true of time-
1252 series of wood and sediment in both reaches, and correlations between middle- and
1253 lower-reach time-series of wood and sediment were also high, suggesting a first-order
1254 control on large wood and sediment due to upstream hydrology and wood and sediment
1255 supply. An increase in wood in the lower reach, despite the destruction of several
1256 engineered logjams during dam removal and subsequent channel adjustment, has likely
1257 been beneficial for salmon restoration, mediated through pool scouring and bar
1258 building.

1259 These results not only provide evidence of the substantial changes to the areal
1260 distribution of floodplain wood and sediment resulting from dam removal, but also
1261 provide evidence that these wood-related changes varied spatially over the study area
1262 reaches and subreaches. While novel, these observations and the understanding gained
1263 from them may be expanded with the future processing and release of dozens more
1264 aerial surveys of the Elwha River that will fill gaps and expand the time-series data.

Future application of these methods has the potential to advance the incomplete theoretical understanding of the dynamics of wood in rivers, such that improvements are made in our abilities to predict the two-way feedbacks of water, sediment, wood, vegetation, and the evolving channel morphology (Swanson et al., 2021). These advances may be possible because of measurements of large wood over meaningful scales are now possible with deep learning techniques such as those shown here. Automation through remote sensing and deep learning is no substitute for field insight, accuracy, or precision, however, and vegetation canopy and volumetric measurements of wood will remain challenging from aerial orthoimagery alone. Thus, hybrid approaches of remote sensing and field observations continue to be necessary. Automation does provide the benefits of near complete coverage and consistent measurements over any scale. In the future, Machine Learning has potentially limitless application in the science of large wood if detailed field-based and/or multi-modal measurements of wood characteristics can be paired with coincident data such as imagery and high-resolution digital elevation products, to produce models that can identify specific features of interest over large scales.

In conclusion, our approach to evaluating wood dynamics in rivers using remote sensing and image segmentation using deep learning has been designed to scale to other similar river systems with similar imagery. We therefore hope our open-source image segmentation software, publicly available models, and datasets will stimulate further application and research into simultaneous large wood dynamics in other river catchments, as well as to facilitate further research on the morphodynamics and environmental restoration of the Elwha River.

Open Research

Image segmentation models are available from Buscombe (2023a) that work with the workflows detailed by Buscombe and Goldstein (2022). All data are publicly and freely available in Ritchie et al., (2018) and Buscombe (2023b), except the time-series

1293 measurements of alluvial bars in the lower reach made by the Lower Elwha Klallam
1294 Tribe.

1295

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